

Deep Learning for teens

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BERT

- ▶ What is BERT
 - ▶ Think of BERT as a model that **reads text like a human** — not just left-to-right, not just right-to-left, but **both directions at the same time**.

Transformers

- ▶ Transformers got two components
 - ▶ Encoder
 - ▶ Converts whole sentence to contextual embedding
 - ▶ Decoder
 - ▶ We give the embedding to decoder to do something for us like text prediction, translating sentence
- ▶ BERT has only encoder part
- ▶ GPT has only decoder part

Transformers

- ▶ Query: You type a Search for a topic.
- ▶ Key: The app looks at the Profiles of other users.
- ▶ Value: You read the Posts (content) from the users that matched your search.

Transformers

- ▶ Query
 - ▶ Need helps to write an essay
- ▶ Attention heading

Model Type	Architecture	Role	Example Models
Encoder-Only	Only the "Reader"	Understanding (Search, Sentiment, Grammar)	BERT, RoBERTa
Decoder-Only	Only the "Writer"	Creation (Chatting, Writing, Coding)	GPT-4, Gemini, Llama, Claude
Encoder-Decoder	Both together	Translation (Read French $\rightarrow$ Write English)	T5, BART, Google Translate

Homework

- ▶ Your mission is to use the Movie Critic dataset, the BERT Tokenizer, and the BERT Model to perform Sentiment Analysis. You will discover how the model uses context to understand whether a review is positive or negative